

HOLY CROSS COLLEGE (AUTONOMOUS), TIRUCHIRAPPALLI - 620 002

COE STANDARD QUESTION BANK IMPORT TEMPLATE (OBE / BLOOM TAXONOMY)

Course Code: U23CS1ALT01 | Title: Data Structures & Algorithms | Language: TA | Canonical 26 Fields

Q.No	Section	Type	Unit	K/CO	Marks	Question Prompt & Options	Correct Key	Answer Text / Solution	Specifics (Passage/Match/Reason)
1	Section A	MCQ	U1 1.1	K1 CO1	1	Which of the following data structures follows the Last-In-First-Out (LIFO) principle? A) Queue B) Stack C) Linked List D) Binary Tree	B	Stack operates strictly on Last-In-First-Out (LIFO) ordering.	Basic recall MCQ
2	Section A	MATCH	U1 1.1	K2 CO1	1	Match the fundamental data structures in Column A with their primary operational properties in Column B: A) (1)-(c), (2)-(a), (3)-(b), (4)-(d) B) (1)-(a), (2)-(c), (3)-(b), (4)-(d) C) (1)-(b), (2)-(a), (3)-(d), (4)-(c) D) (1)-(c), (2)-(d), (3)-(a), (4)-(b)	A	1-c (Stack/LIFO), 2-a (Queue/FIFO), 3-b (Binary Tree/Hierarchical), 4-d (Graph/Non-linear network)	Match: 1. Stack 2. Queue 3. Binary Tree 4. Grap... vs a. First-In-First-Out (FIFO) discipline ...
3	Section A	ASSERTION_REASON	U1 1.2	K2 CO1	1	Read the assertion and reason statements carefully and select the correct option: A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A. B) Both Assertion (A) and Reason (R) are true, but R is NOT the correct explanation of A. C) Assertion (A) is true, but Reason (R) is false. D) Both Assertion (A) and Reason (R) are false.	A	Binary search requires sorted array elements to perform divide-and-conquer splitting.	Assertion: Assertion (A): Binary search can only be performed on a sort... Reason: Reason (R): Binary search eliminates half of the search spac...
4	Section A	FILL_BLANK	U1 1.2	K1 CO1	1	The time complexity to access any element in an array using an index offset is _____.	O(1)	O(1) constant time due to direct pointer arithmetic.	Fill in the blank
5	Section A	VSA	U1 1.2	K1 CO1	2	Define time complexity of an algorithm and state the Big-O asymptotic notation for linear search.	Time complexity measures execution growth rate; linear search is O(n).	Time complexity quantifies the amount of running time required by an algorithm as a function of the length of the input	Very Short Answer
6	Section B	PARAGRAPH	U2 2.1	K2 CO2	5	Explain the memory representation of single and multi-dimensional row-major arrays with address calculation formulas.	Base address computation formula: $\text{Loc}(A[i][j]) = \text{Base} + w * [(i - \text{LB1}) * N2 + (j - \text{LB2})]$	1. Single-dimensional array: $\text{Loc}(A[i]) = \text{Base}(A) + w * (i - \text{LB})$. 2. 2D Row-Major formula: $\text{Loc}(A[i][j]) = \text{Base}(A) + w * [$	Paragraph / Short Answer

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7	Section B	EITHER_OR	U2 2.2	K3 CO2	5	(a) Describe the array implementation of Circular Queue with wrap-around boundary condition algorithms. OR (b) Explain the algorithm to convert an Infix expression into Postfix using a stack with an illustrative dry-run table.	Circular Queue front/rear modulo logic OR Infix to Postfix Shunting-yard algorithm.	Circular queue front/rear calculation: rear = (rear + 1) % MAX, (front + 1) % MAX OR Infix to Postfix stack precedence t	Choice: Q07_CHOICE
8	Section C	PASSAGE	U3 3.1	K4 CO3	8	Read the system architectural scenario below and answer the analytical questions that follow:	Analytical solutions addressing tree balancing, height bounds, and indexing optimization.	1. Skewed insertion causes BST degenerate worst-case complexity O(n). 2. AVL strict balance factor BF = Height(Left) - H	Passage: «In a mission-critical financial trading engine, an unbalanced binary search tre...
9	Section D	ESSAY	U4 4.1	K5 CO4	10	Elaborate on AVL Tree re-balancing rotations (LL, RR, LR, RL) with balance factor criteria and construct a step-by-step balanced AVL tree for the insertion sequence: 15, 20, 24, 10, 13, 7, 30, 36, 25.	Complete step-by-step AVL insertion with balance factors and rotation diagrams.	1. Balance Factor theorem $BF = h_L - h_R \in \{-1, 0, 1\}$. 2. Detailed Single Rotations (LL, RR) and Double Rotations (LR, R	Comprehensive Essay